Green Roof Implementation for Flood Mitigation

Final Project Proposal

LA 5580 | Web Mapping and Data Visualization

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Project Title

Green Roof Implementation for Flood Mitigation: A GIS-Based Suitability Analysis of the Kisena Park Area, Queens, New York

Abstract

This project conducts a GIS-based suitability analysis to identify optimal rooftops for green roof installation in the Kisena Park neighborhood of Queens, New York. Using spatial data on building characteristics, solar radiation, proximity to water, and structural attributes, this study evaluates which buildings are best positioned to support green roof systems as a nature-based solution to urban flooding. Results will be presented through interactive visualizations in Tableau, highlighting suitability patterns across the neighborhood and providing actionable recommendations for urban planners and policymakers seeking to enhance stormwater management and climate resilience in flood-prone communities.

Overview

The Kisena Park area of Queens, New York, is one of the most flood-vulnerable neighborhoods in New York City. Its low-lying topography and proximity to water bodies create conditions that amplify flood risk during heavy rainfall and storm surge events. According to the New York City Department of Environmental Protection, the area has recorded over 200 street flooding incidents in the past five years. Much of the neighborhood sits only 2 to 4 feet above mean sea level, rendering it particularly susceptible to both tidal flooding and intense precipitation events exacerbated by climate change.

Green roofs are an established nature-based solution (NBS) that reduce stormwater runoff by retaining rainfall on vegetated roof surfaces, thereby decreasing peak flow volumes entering urban drainage systems. However, not all buildings are equally suited for green roof installation. Structural load capacity, roof area, age, current use, solar exposure, and access to irrigation sources all influence feasibility and effectiveness.

This project will apply a multi-criteria suitability analysis using GIS to evaluate buildings within the Kisena Park area against these criteria. The spatial analysis will culminate in a series of suitability maps identifying high-priority rooftops for green roof deployment. Results will be visualized and communicated through Tableau dashboards, enabling planners, local government, and community stakeholders to explore and act on the findings.

The overarching spatial question guiding this project is: Which buildings in the Kisena Park area are most suitable for green roof implementation to effectively reduce stormwater runoff and flooding risk?

Technology

The following technologies will be utilized to complete this project:

- ArcGIS Pro: Primary platform for spatial data processing, multi-criteria suitability analysis, and production of final GIS maps.
- Tableau: Data visualization and interactive dashboard creation for communicating suitability scores and spatial patterns.
- R: Data collection and Leaflet map visualization.
- Microsoft Excel: Data cleaning, attribute table management, and pre-processing of tabular datasets prior to GIS integration.
- Github: Hosting webmaps in online portfolio.

Data Sources

The following datasets will be used in this analysis:

- NYC Department of City Planning: Building Footprint Dataset: Rooftop dimensions, building age, land use classification, and number of floors.
- NYC Open Data: Building Information: Building height, structural type, and construction materials to assess load-bearing feasibility.
- USGS National Elevation Dataset (NED): Digital Elevation Model (DEM) for topographic analysis and flood-risk delineation.
- NYC DEP Flooding: Incident Reports: Historical Street flooding event data to validate spatial flood-risk patterns.

Process and Methods

- **Data Acquisition & Cleaning:** Download all datasets from respective portals; standardize projections to NY State Plane (EPSG: 2263); remove incomplete or null records; join tabular building attribute data to spatial footprints.
- **Criteria Layer Development:** Generate individual suitability layers for each criterion: (1) rooftop area, (2) building height, (3) building age, (4) building use type, (5) solar radiation index, and (6) proximity to water infrastructure.
- **Reclassification & Scoring:** Reclassify each criterion layer on a standardized 1–5 suitability scale using defined thresholds (e.g., larger rooftop area = higher suitability score).
- **Weighted Overlay Analysis:** Apply weighted multi-criteria evaluation (WMCE) in ArcGIS Spatial Analyst, assigning weights to each criterion based on literature-informed importance. Produce a composite suitability score for each building footprint.
- **Map Production:** Generate final suitability maps in ArcGIS Pro displaying composite scores across all buildings in the study area.
- **Tableau Visualization:** Build interactive Tableau dashboards presenting criterion-bycriterion breakdowns, overall suitability maps, and comparative building profiles for stakeholder communication.
- **GitHub Embedding:** Embed final maps in Github to display in online portfolio.

Inspiration

The following resources informed the conceptual and methodological design of this project:

1. Kim, H., Oh, K., & Lee, D. (2021). An algorithm for identifying suitable locations for green roofs and roadside trees using GIS spatial analysis: A case study of Seoul. Urban Forestry & Urban Greening, 58, 126-136. https://doi.org/10.1016/j.ufug.2021.126136 The objective of the research is to develop a robust algorithm that utilizes GIS spatial analysis to identify suitable locations for the implementation of green roofs and roadside trees in urban environments, specifically focusing on the city of Seoul. The study aims to enhance urban ecosystem services in response to climate change by assessing building characteristics and sidewalk conditions, ultimately facilitating urban greening initiatives that contribute to temperature reduction and improved environmental quality.

2. Grunwald, L., Heusinger, J., & Weber, S. (2017). A GIS-based mapping methodology to assess urban green roof ecosystem services and identify priority green roofs in Braunschweig, Germany. Urban Forestry & Urban Greening, 24, 166-176. https://doi.org/10.1016/j.ufug.2017.02.006 The objective of the research is to develop a GISbased mapping methodology to assess the spatial variation of urban green roof ecosystem services (UESS) in Braunschweig, Germany, and to identify priority green roofs (PRIOGR) for implementation. The study evaluates four key categories of UESS; thermal urban climate regulation, air quality improvement, stormwater retention, and biodiversity enhancement by analyzing geospatial datasets and classifying buildings based on their suitability for green roofs. This approach aims to support urban planning strategies and promote the integration of green infrastructure in dense urban environments.

3. Sultana, N., Sharifi, A., Haque, M. N., & Aghaloo, K. (2021). An integrated GIS-based multicriteria decision-making approach for evaluating rooftop suitability for agriculture in Dhaka, Bangladesh. Urban Agriculture & Regional Food Systems, 6(1), 1-13. https://doi.org/10.1017/auf.2021.1 The objective of the research is to develop an integrated GIS-based multi-criteria decision-making (MCDM) approach to evaluate the suitability of rooftops in Dhaka, Bangladesh, for agricultural use. By employing the Best-Worst Method (BWM), the study identifies and weighs seven key criteria, including rooftop area, building age, and sunlight availability to assess rooftop agricultural potential. The findings aim to inform policymakers and urban planners about the most suitable locations for rooftop agriculture, thereby enhancing climate resilience, food security, and sustainable urban development in Dhaka.

Potential Challenges

- Data Completeness: Building attribute data (particularly structural load capacity) may be incomplete for older structures, necessitating the use of proxy variables such as building age and construction type.
- Projection and Alignment: Integrating raster datasets (e.g., solar radiation, DEM) with vector building footprint data requires careful CRS alignment and resampling to avoid spatial misregistration.

- Weighting Subjectivity: Assigning weights in the multi-criteria evaluation introduces methodological subjectivity. This will be addressed by grounding weight assignments in peerreviewed literature and conducting a sensitivity analysis.
- Tableau Integration: Connecting and visualizing complex spatial outputs in Tableau may require intermediate export steps (e.g., exporting feature attribute tables to CSV or using Tableau's spatial file connector).

Timeline for Completion

- **Week 1-2:** Data acquisition, download, and initial cleaning; CRS standardization; attribute table joins. Development of individual suitability criterion layers; reclassification and scoring.
- **Week 3-4:** Weighted overlay analysis; composite suitability map generation; flood risk integration. Tableau dashboard development; interactive visualization design and layout. Final report writing, cartographic refinement, and project submission.

AI Usage Disclosure

Artificial intelligence was used in the preparation of this proposal, specifically, Claude. AI was used to assist in formatting proposal structure.AI was used solely as a writing and organizational aid; all substantive analytical and design decisions reflect the author's own intellectual contribution.